Process Characterization Plan

A-Mab Drug Substance — Production Bioreactor (Step 3) — PCP-003

Process Development / Manufacturing Science & Technology (MSAT)

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Author	Process Development / MSAT	—	—	—
Reviewer	Analytical Development	—	—	—
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Abbreviations

Table 2: Abbreviations used in this plan.

Abbreviation	Expansion
AMV	analytical method validation
CHO	Chinese hamster ovary
CPP	critical process parameter
CQA	critical quality attribute
DoE	design of experiments
DS	drug substance
GPP	general process parameter
HCP	host cell protein
HILIC	hydrophilic interaction liquid chromatography
icIEF	imaged capillary isoelectric focusing
IgG1	immunoglobulin G, subclass 1
KPP	key process parameter
MSAT	Manufacturing Science and Technology
NOR	normal operating range
PAR	proven acceptable range
pCO2	partial pressure of dissolved carbon dioxide
PD	Process Development
PPQ	process performance qualification
qPCR	quantitative polymerase chain reaction
QA	Quality Assurance
RSM	response surface methodology
SEC	size exclusion chromatography
SOP	standard operating procedure
UPLC	ultra performance liquid chromatography
WC-CPP	well controlled critical process parameter

1 Purpose and scope

The production bioreactor is Step 3 of the A-Mab drug substance process and the step at which the glycan, charge variant and aggregate attributes of the molecule are formed. Glycosylation is completed in the Golgi before the antibody is secreted. Protein A binds the Fc region of the antibody, and the cation and anion exchange steps separate on net surface charge, so none of the three retains one glycoform in preference to another. The glycoform distribution of the drug substance is therefore decided in the bioreactor. Aggregate and acidic charge variants are also formed here, and the purification train can reduce that burden but cannot reverse it. This plan defines the studies that will characterize the step and the analyses that will be run on the data they produce.

The plan covers the 9 process parameters of the production bioreactor listed in Table 6.1 and the 7 quality attributes of the 10 in the drug substance register that this step forms (Table 4.1). The studies will be executed on the qualified scale-down bioreactor model described in §5.1, operated at conditions equivalent to the 15,000 L commercial production vessel. Each parameter will be studied over a range wider than the range in which the process is expected to run, so that the study bounds behaviour outside the intended operating region as well as inside it.

Three activities are outside this plan. The seed expansion steps that precede the production bioreactor are characterized under PTP-001 and are not repeated here. Harvest and clarification, which separate the cells from the culture fluid by centrifugation and remove the remaining debris by depth filtration, are covered by PCP-004. Confirmation of the ranges at commercial scale belongs to Stage 2 and will be performed during process performance qualification at the receiving site (U.S. Food and Drug Administration 2011). The studies planned here are Stage 1 process design studies, and the claims they support are bounded by the scale-down model on which they are run.

The executed studies, their results and the parameter classification that follows from them will be reported in PCR-003.

2 Objectives

The objectives of the studies planned here are the following.

(i) To quantify the relationship between each studied parameter and each of the 5 modelled responses over the characterization ranges given in Table 6.1, and to identify the interactions between parameters.

(ii) To define the multivariate region of the well controlled parameters over which every modelled attribute in Table 4.1 is predicted to meet its acceptance criterion.

(iii) To establish a proven acceptable range for each parameter against each attribute whose fitted model contains a significant term in that parameter, and to confirm that the normal operating range in Table 6.1 lies inside that range.

(iv) To provide the demonstrated effects and the operating risk from which each parameter will be classified as a critical process parameter, a well controlled critical process parameter, a key process parameter or a general process parameter under SOP-4001.

(v) To provide the process understanding and the capability estimate that justify the Stage 2 operating ranges and this step's contribution to the control strategy.

Objectives (i) to (iii) are met by the studies described in §6 and judged against the criteria in §7 and §8. Objectives (iv) and (v) are met by the analyses of §5.5 and reported in PCR-003.

3 Related documents

Table 3.1 lists the documents this plan depends on and the document it feeds. RA-001 set the scope of the study by ranking the parameters of every step and assigning each one a study type. PCMP-001 states the campaign level approach that this plan instantiates for the production bioreactor. The executed studies will be reported in PCR-003, which rolls up into PCMR-001.

Table 3.1: Documents related to this plan.

Document ID	Title	Relationship
PTP-001	Process Transfer Plan (A-Mab Drug Substance)	Parent — transfer scope
RA-001	Pre-Characterization Process Risk Assessment (A-Mab Drug Substance)	Parent — risk basis for study scope
PCMP-001	Process Characterization Master Plan (A-Mab Drug Substance)	Parent — master plan
PCR-003	Process Characterization Report (Production Bioreactor (Step 3))	Sibling — paired document
PCMR-001	Process Characterization Master Report (A-Mab Drug Substance)	Rolls up into

Table 3.2 lists the controlled procedures and the validated analytical methods that apply to execution and analysis. The operation of the step follows SOP-2003, the preparation of medium and feed follows SOP-2004, the qualification of the scale-down model follows SOP-1001, and the design, randomization and statistical analysis of the multivariate studies follow SOP-1002. Parameter classification follows SOP-4001.

Table 3.2: Controlled procedures and validated analytical methods for this study.

Reference	Title	Type
SOP-1001	Qualification of Scale-Down Models	SOP
SOP-1002	Design, Execution and Statistical Analysis of DoE Studies	SOP
SOP-2003	Operation of the Production Bioreactor (Fed-Batch)	SOP
SOP-2004	Cell-Culture Media and Feed Preparation	SOP
SOP-3010	Released N-Glycan Mapping by 2-AB HILIC-UPLC	SOP
SOP-3011	Size-Variant Analysis by SEC-HPLC	SOP
SOP-3012	Host-Cell-Protein Quantitation by ELISA	SOP

Reference	Title	Type
SOP-3013	Charge-Variant Analysis by icIEF	SOP
SOP-3014	Residual Host-Cell DNA by qPCR	SOP
SOP-4001	Process-Parameter Classification and Control-Strategy Definition	SOP
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

4 Prior knowledge and quality risk basis

4.1 Unit-operation description and prior knowledge

The production bioreactor is a fed-batch culture of a recombinant CHO cell line that secretes A-Mab into the medium over a culture of about 17 days. Culture pH, temperature, dissolved oxygen, dissolved carbon dioxide and osmolality are held at set-points by the vessel control system throughout that period. Each of them changes the specific growth rate of the cells and the rates of the Golgi enzymes that complete the post-translational modification of the antibody. The harvested pool is the integral of everything the culture secreted during the run, so a longer culture weights the pool toward antibody made after the nucleotide sugar donors have been drawn down and after viability has begun to fall.

The platform on which A-Mab is produced has been used for the humanized IgG1 antibodies X-Mab, Y-Mab and Z-Mab, and for the A-Mab clinical and engineering campaigns (CMC Biotech Working Group 2009). Cell line construction, basal medium, feed strategy and the control scheme for pH, temperature and dissolved oxygen are common to those products. The same host cell line carries the same complement of glycosyltransferases and glycosidases, and the same basal medium and feed supply the same nucleotide sugar precursors, so a parameter that lowered galactosylation in those products lowers it in A-Mab through the same enzymes. The directions recorded on the platform are therefore carried into this study as expectations, and the magnitudes are not.

The glycan attributes are formed by the enzymes of the Golgi and by the supply of the nucleotide sugar donors. Core fucose is added by a fucosyltransferase from GDP-fucose, so the fraction of antibody that leaves the cell without core fucose rises when the activity of that enzyme falls, when GDP-fucose is depleted, or when the glycan spends less time in the Golgi before it is secreted. Terminal galactose is added by a galactosyltransferase from UDP-galactose, and both that donor and the manganese cofactor of the enzyme are depleted late in a fed-batch culture. Galactosylation is therefore expected to fall as the culture ages and the harvested pool accumulates late-culture antibody. High mannose species are those on which trimming and the later additions never proceeded, so their level is expected to rise when transit through the Golgi is fast relative to the processing enzymes or when those enzymes are inhibited. The fucosyltransferase, the galactosyltransferase and the mannosidases draw on nucleotide sugar pools of a fixed size and work at a single Golgi luminal pH, so raising culture pH or temperature changes the rate of all of them at once. The amount by which raising one parameter moves a glycan attribute is therefore expected to depend on the levels of the others.

Aggregate and acidic charge variants are formed in the culture fluid after the antibody is secreted. Aggregate arises from antibody that partially unfolds and then associates while it sits at culture temperature for days, so a higher titer, a longer culture and a culture pH closer to the isoelectric point of the antibody each raise the aggregate level in the harvest. Raising culture pH across the range in Table 6.1 lowers the net positive charge the antibody carries, which weakens the electrostatic repulsion between molecules and raises the rate at which they associate. Acidic charge variants are mainly the products of asparagine deamidation, together with sialylated glycans and glycated lysines. The rate of asparagine deamidation rises with pH and with temperature, and the time available for it rises with the culture duration. Sialic acid is transferred by the sialyltransferases from CMP-sialic acid, and glycation adds glucose to lysine residues without an enzyme at a rate set by the residual glucose in the medium. These three routes do not respond to a given parameter in the same direction, so the net direction in which that parameter moves the acidic variant level is not predictable from the chemistry alone.

Host cell protein and residual DNA enter the harvest when cells die and lyse. Viability falls late in a fed-batch culture, so a longer culture carries a larger burden of both into harvest. Protein A binds the Fc region of the antibody and lets host cell protein and DNA pass into the flow through and the wash. The cation exchange step binds the antibody and washes further host cell protein away, and the anion exchange step binds DNA and host cell protein while the antibody flows through.

A higher culture pH raises intracellular pH and with it the Golgi luminal pH. The glycosyltransferases and the mannosidases each work fastest at a different luminal pH, so a shift changes their rates by different amounts and moves the glycan distribution. A higher culture pH also raises the rate of asparagine deamidation and lowers the net positive charge of the antibody in the culture fluid.

A higher culture temperature raises the specific growth rate and the rate of every enzymatic and chemical reaction in and around the cell. A cooler culture grows more slowly and stays productive for longer, and a warmer one deamidates and aggregates faster in the culture fluid.

Dissolved carbon dioxide equilibrates with bicarbonate and lowers cytosolic pH, so raising it moves the Golgi enzymes in the same direction as lowering culture pH. Holding culture pH at its set-point against that acidification consumes more base, which raises osmolality as bicarbonate and sodium accumulate.

Osmolality above the set-point draws water out of the cell, slows the specific growth rate and raises the amount of antibody each surviving cell makes. It also changes the size of the intracellular nucleotide sugar pools, and those are the pools that culture pH and temperature change, so its effect on a glycan attribute is expected to depend on the levels of those two.

A longer culture lowers viability at harvest and raises the fraction of the pool secreted after the UDP-galactose and manganese supply has fallen. It therefore lowers galactosylation and raises the host cell protein and DNA burden carried into harvest.

Dissolved oxygen supplies the terminal electron acceptor for oxidative phosphorylation, so a lower level slows the rate at which the cell regenerates ATP. The culture is not oxygen limited at either edge of the range in Table 6.1, so neither the specific growth rate nor the glycan distribution is expected to move measurably across it.

A higher initial viable cell concentration brings the culture to its peak viable cell density earlier and leaves more days in the productive phase before harvest, which raises titer. A higher basal medium concentration and a larger nutrient feed raise the supply of glucose, amino acids and nucleotide sugar precursors, which raises growth and titer. Across the ranges in Table 6.1 the feed replenishes those precursors faster than the cells consume them, so none of the three is expected to move a glycan attribute at a fixed culture age.

4.2 Quality attributes in scope

Table 4.1 lists the attributes this step forms, with the acceptance criterion for the drug substance, the criticality level and the Tool #1 score assigned in RA-001. Criticality was scored as the product of the impact of the attribute on safety or efficacy and the uncertainty in that impact, and an attribute scored high or very high on that continuum is called critical (CMC Biotech Working Group 2009; International Council for Harmonisation 2023).

Table 4.1: Quality attributes formed at the production bioreactor, with drug substance acceptance criteria and criticality.

CQA	Category	Acceptance	Criticality	Tool #1
Afucosylation	glycosylation	2–13 % of glycans	H	60
Galactosylation (total %Gal)	glycosylation	10–40 % (G1+G2 equiv.)	H	48
High Mannose	glycosylation	3–10 % of glycans	VH	80
Aggregates (HMW)	size variant	0–5 % HMW (SEC)	H	60
Acidic charge variants (deamidation)	charge variant	18–40 % (CEX/icIEF)	VL	4
Host Cell Protein (HCP)	process impurity	0–100 ng/mg	M-H	36
Residual DNA	process impurity	0–0.001 ng/dose	VL	6

High mannose carries the highest Tool #1 score of the attributes in Table 4.1. High mannose species are cleared from the circulation faster than the fully processed glycoforms, so a raised level shortens exposure. Afucosylation and galactosylation are scored high because the mechanism of action of A-Mab is primarily antibody dependent cellular cytotoxicity. Removing the core fucose from the Fc glycan raises the affinity of the antibody for the Fc gamma receptor IIIa on the effector cell, and a higher affinity raises the cytotoxicity the bound cell delivers. Aggregate is scored high for its immunogenic potential. The acidic charge variants and residual DNA are scored very low, and host cell protein between moderate and high.

5 of the 7 attributes will be measured as responses of the designed experiments. They are the three glycan attributes, the acidic charge variants and the high molecular weight aggregate. Host cell protein and residual DNA will be measured on the clarified harvest of every run (§5.4), but they are not modelled responses of this study. Protein A capture and the two polishing steps remove most of both, and the ranges over which they are controlled are established in PCR-005, PCR-007 and PCR-008.

4.3 Risk-based prioritization of parameters

RA-001 ranked every parameter of this step on two axes, the potential impact of the parameter on a quality attribute and the potential for the parameter to interact with another parameter, and filtered the ranked list into study types (CMC Biotech Working Group 2009). 5 parameters were assigned to the multivariate design and 4 to univariate assessment. Every parameter of this step is studied. No later step adds or removes a glycan, so a parameter that moves a glycan attribute here cannot be compensated further down the train.

The 5 multivariate parameters are Culture pH, Culture temperature, Culture duration, Dissolved CO2 (pCO2) and Osmolality. Each of them changes Golgi luminal pH or the size of the nucleotide sugar pools, and the glycosyltransferases draw on both, so the amount by which one of them moves a glycan attribute is expected to depend on the levels of the others. A design that varies them together is the only design that estimates that dependence. The 4 univariate parameters are Dissolved oxygen, Initial viable cell conc., Basal medium concentration, Nutrient feed-1 volume. Each of them raises or lowers the number of viable cells and the amount of antibody the culture makes, and the nutrient feed holds the nucleotide sugar precursors in excess across the ranges studied, so none of them is expected to change the rate of a Golgi enzyme at a fixed culture age. A study that varies one at a time therefore supports the range each of them needs.

The ranges to be studied are the knowledge space edges of each parameter and are wider than the ranges in which the process is expected to run. Across the 9 parameters the characterization range is 1.5 to 2.5 times the width of the normal operating range. The wider ranges are used for two reasons. They bound the response of each attribute outside the region the process should occupy, which is what allows the severity of an excursion to be assessed. They also support later movement within the design space without new studies (CMC Biotech Working Group 2009). No behaviour is predicted outside the ranges in Table 6.1.

5 Materials and methods

5.1 Scale-down model and its qualification

The studies will be executed in bench-scale stirred tank bioreactors of the same design class as the commercial production vessel, with equivalent mixing, aeration and mass transfer characteristics and the same control scheme for pH, temperature, dissolved oxygen and nutrient addition. Scale independent parameters, which include culture pH, temperature, dissolved oxygen, initial viable cell concentration and culture duration, will be operated at the commercial set-points in Table 6.1. Scale dependent parameters, which include agitation rate, gas flow rates, headspace pressure and working volume, cannot be held equal across scales and will be set so that process performance at small scale matches performance at full scale (CMC Biotech Working Group 2009).

Dissolved carbon dioxide is the parameter for which that distinction matters most. Carbon dioxide accumulates in a large vessel because the liquid height raises the hydrostatic head and lowers the volumetric mass transfer coefficient for carbon dioxide stripping. A bench-scale vessel strips carbon dioxide faster at the same gas flow per unit volume, so it reaches a lower dissolved carbon dioxide level under the same nominal conditions. Dissolved carbon dioxide is therefore controlled to a set-point in the scale-down model and is studied as a factor in its own right, over a range that spans the levels the commercial vessel reaches.

Qualification of the model will follow SOP-1001. Replicate runs at the set-point conditions of Table 6.1 will be compared against the at-scale data available from the A-Mab engineering and clinical campaigns, on the input attributes of the culture and on the output attributes of the harvest. The comparison covers the viable cell density profile, viability at harvest, titer and the 5 quality responses of Table 4.1. The model will be considered qualified when that comparison meets the criterion in §7, and the qualification record will be referenced in PCR-003.

The centre point runs of both designs are executed at the same set-point conditions as the qualification runs. They therefore carry the run to run variability of the model together with the variability of the analytical methods, and they provide the pure error term against which lack of fit is tested (§5.5).

5.2 Cell line, media and culture operation

Cells will be thawed from the working cell bank and expanded through the seed train to the inoculation density given in Table 6.1. The production bioreactor will be operated in fed-batch mode under SOP-2003, with a single nutrient feed added on the platform schedule. Culture pH will be controlled by carbon dioxide addition and base addition, temperature by the vessel jacket, and dissolved oxygen by sparging and agitation. Basal medium and feed will be prepared under SOP-2004 from the platform formulations. The culture will be harvested at the culture duration set for the run.

Inputs that are controlled to a platform specification are identity and quality controlled on receipt and are not characterized in this study. They include the medium and feed lots, the gases and the base. Their specifications were set on the platform and are unchanged for A-Mab, so the variability they contribute is already bounded by the specification and by the release testing applied to each lot. Where a lot of feed or medium is used across more than one design, its identity will be recorded with the run so that a lot effect can be examined if the data suggest one.

5.3 Analytical methods

Table 5.1 lists the validated methods that will generate the study data. The released N-glycan map by 2-AB HILIC-UPLC measures afucosylation, galactosylation and high mannose from a single preparation, so the three glycan responses of any one run are not independent measurements. Size exclusion chromatography measures the high molecular weight aggregate. Imaged capillary isoelectric focusing measures the acidic charge variants. The host cell protein ELISA and the residual DNA qPCR measure the two impurity attributes on the clarified harvest.

Table 5.1: Validated analytical methods and the responses they measure.

Reference	Title	Type
AMV-3010	N-Glycan Map (2-AB HILIC-UPLC)	Method validation
AMV-3011	Size-Variants (SEC-HPLC)	Method validation
AMV-3012	Host-Cell Protein ELISA	Method validation
AMV-3013	Charge Variants (icIEF)	Method validation
AMV-3014	Residual DNA (qPCR)	Method validation

Each method will be used within its validated range and under its current validation report. No method will be modified for this study. Method precision enters the study twice. It contributes to the variability observed between the centre point replicates, and it therefore enters the pure error term used to test lack of fit. It also puts a floor under the smallest change in a response that the study can resolve, so an effect of the same size as the method precision will be reported as detected but not distinguishable from analytical variability. Method performance detail is held in the validation reports listed in Table 5.1 and is not reproduced here.

5.4 Sampling plan

Each run will be sampled at harvest, and the harvest sample will be assayed for the 5 modelled responses. The clarified harvest of each run will be assayed for host cell protein and residual DNA. Samples will be taken by the procedure in SOP-2003 and will be handled and stored identically across runs, because a difference in hold time before assay would appear in the data as a run effect.

The culture will be monitored through the run for viable cell density, viability and the principal metabolites, under SOP-2003. These measurements describe how the culture progressed and support the investigation of any run that fails an operating criterion. They are not responses of the designs and no model will be fitted to them.

The screening design includes 3 centre point runs and the response surface design includes 4, all executed at the set-points in Table 6.1. The replicates are distributed through the execution order, so that any drift in the medium and feed lots or in the analytical methods across the campaign enters the pure error term.

5.5 Statistical methods

Every factor will be coded so that the set-point is 0 and the edges of the characterization range are −1 and +1. Coding puts the factors on a common scale, so a coefficient measures the change in the response produced by moving from the set-point to the edge of the range, and coefficients for different factors may be compared directly (CMC Biotech Working Group 2009). An effect will be reported as twice the coded coefficient, which is the change in the response across the full characterization range of that factor.

The screening data will be fitted by ordinary least squares to a model containing the main effects and the two factor interactions. That model has 16 terms and the design has 19 runs, so it is near saturated and leaves 1 degree of freedom for lack of fit against 2 degrees of freedom of pure error. The screening analysis will therefore be read as identifying which factors and which interactions are active. It will not be used to predict a response, to define an operating region or to set a range. The response surface model is the predictive model for all three of those purposes.

The response surface data will be fitted to the full quadratic model in the 4 factors carried forward, which has 15 terms in 28 runs. Model adequacy will be judged on the coefficient of determination, the adjusted coefficient of determination, the predicted coefficient of determination computed from the prediction error sum of squares, the overall F test, and an analysis of variance that partitions the residual into lack of fit and pure error. The lack of fit test uses the 4 centre point replicates and has 10 and 3 degrees of freedom. Terms will be assessed at a significance level of $p < 0.05$. Residual diagnostics will be examined for every response that is modelled, as residuals against predicted values, a normal quantile plot and predicted against measured values.

A factor with no term significant at $p < 0.05$ for a given response will be reported as showing no detectable effect on that response over the range studied. The statement holds over the characterization range in Table 6.1 and no further. It also holds only down to the smallest effect the design and the analytical method can resolve, which the centre point replicates measure directly. An effect that is significant but small will be reported with the predicted change across the range beside it, so that statistical significance and practical significance are separated in the report (CMC Biotech Working Group 2009).

Process capability at commercial scale will be estimated by Monte-Carlo simulation of 2,000 batches. Each simulated batch draws every parameter from within its normal operating range and evaluates the fitted response surface models together with the variability observed in the study. Capability will be reported as a one-sided Cpk for each attribute, against the criterion that attribute must meet, with the margin to the limit stated. The variability drawn on in the simulation is the variability of the scale-down model, which does not carry the contribution the commercial vessel makes to run to run variation. The capability reported will therefore be confirmed against the commercial-scale batches of Stage 2.

6 Study design

6.1 Factors, ranges and study type

Table 6.1 gives the parameters to be studied, their set-points, the range each will be studied over, the normal operating range and the study type assigned in RA-001. The set-point column is the target the commercial process will be operated at. The normal operating range is the band within which routine operation is expected to hold the parameter, and the range studied is wider than that band for every parameter.

Table 6.1: Parameters to be studied, with set-points, characterization ranges, normal operating ranges and study type.

Parameter	Unit	Set-point	Range studied	NOR	Study type
Culture pH	pH	6.85	6.6–7.1	6.75–6.95	multivariate
Culture temperature	°C	35	34–36	34.5–35.5	multivariate
Dissolved CO2 (pCO2)	mmHg	100	40–160	70–130	multivariate
Osmolality	mOsm	400	360–440	375–425	multivariate
Culture duration	days	17	15–19	16–18	multivariate
Dissolved oxygen	% sat	50	30–70	40–60	univariate
Initial viable cell conc.	e6 cells/mL	1	0.5–1.5	0.8–1.2	univariate
Basal medium concentration	X	1.2	0.8–1.6	1–1.4	univariate
Nutrient feed-1 volume	% WV	12	9–15	10–14	univariate

Table 6.2 gives the letter codes used for the multivariate factors in the design matrices of Appendices A and B and in the effect tables of PCR-003. The codes are assigned in screening order and are kept for the response surface terms, so that a factor carries the same letter throughout the study.

Table 6.2: Coded factor legend for the multivariate designs.

Code	Factor	Unit	Studied in
A	Culture pH	pH	screening + RSM
B	Culture temperature	°C	screening + RSM
C	Dissolved CO2 (pCO2)	mmHg	screening + RSM
D	Osmolality	mOsm	screening
E	Culture duration	days	screening + RSM

6.2 Screening design

The screening design is a 16 run half fraction of the full 32 run two level factorial in the 5 multivariate parameters, augmented with 3 centre points, for 19 runs in total. The fraction is of resolution V. All 5 main effects and all 10 two factor interactions are therefore estimable clear of one another, and each is aliased only with interactions of three factors or more. The planned matrix is given in Appendix A.

Runs will be executed in randomized order under SOP-1002. Randomization protects the estimates against any drift in the model, in the feed lots or in the analytical methods over the execution period, which would otherwise be confounded with the factor settings.

The design carries no quadratic term, so it cannot describe curvature. A difference between the mean of the centre point runs and the mean of the factorial runs will be read as evidence of curvature in that response, and the response surface design will resolve it. The screening design also cannot separate a two factor interaction from an interaction of three factors, which is the price of running a half fraction. A three factor interaction would mean that the amount by which temperature changes the effect of pH on a glycan attribute itself changes with dissolved carbon dioxide. No such behaviour has been seen on the platform, and the parameters carried into the response surface design are re-estimated there in a design that carries no such alias.

6.3 Response-surface design

The response surface design is a face centred central composite design in 4 of the 5 multivariate parameters. It comprises 16 factorial runs, 8 axial runs and 4 centre points, for 28 runs in total. The axial runs sit on the faces of the cube, at the edges of the characterization ranges in Table 6.1. A rotatable design would place them beyond those edges, at a culture pH and a temperature the vessel control system holds only with difficulty and at which the culture is no longer representative of commercial operation. The planned matrix is given in Appendix B. The design supports the full quadratic model described in §5.5 and is the source of every predictive claim the report will make.

The 4 parameters planned for this design are Culture pH, Culture temperature, Culture duration, Dissolved CO2 (pCO2). The parameter not carried forward is Osmolality, whose characterization range is the narrowest of the 5 relative to its normal operating range, at 1.6 times that width. Its main effect and its interactions with the other four are estimated in the screening design, and its normal operating range is the narrowest region of the five in which routine operation may move.

The subset above is the planned subset. The parameters actually carried into the response surface design will be those the screening analysis identifies as active on the attributes of Table 4.1. If screening identifies a different subset, the design will be amended before execution, the amendment will be approved by the functions listed in Table 1, and it will be reported in PCR-003 with the screening result that prompted it.

6.4 Univariate assessment

4 parameters will be assessed one at a time. They are Dissolved oxygen, Initial viable cell conc., Basal medium concentration, Nutrient feed-1 volume. Each will be run at the low edge and the high edge of its characterization range, with every other parameter of Table 6.1 held at its set-point, and the harvest will be assayed for the 5 responses and for the two impurity attributes.

This design cannot determine whether the effect of one of these parameters depends on the level of another parameter (International Council for Harmonisation 2009). The range it supports for each parameter is therefore conditioned on the other parameters sitting at their set-points, and the report will state that condition where it states the range. The condition is accepted for these 4 parameters because each of them raises or lowers the number of viable cells and the amount of antibody they make, and because the nutrient feed holds the nucleotide sugar precursors in excess across the ranges studied, so none of them changes the rate of a Golgi enzyme at a fixed culture age. If a univariate assessment shows an effect on a quality attribute that the mechanism does not account for, the parameter will be carried into a multivariate study and the plan amended.

7 Acceptance and decision criteria

This section states the rules by which the study will be judged. The rules are fixed before the data exist, and any change to them after execution begins will be recorded as an amendment to this plan.

The scale-down model will be considered qualified when the comparison described in §5.1 shows no statistically significant difference between the scale-down runs and the at-scale data for the compared attributes, at a significance level of $p < 0.05$. If a difference is found, the source of the difference will be investigated before any characterization run is executed, and the model will be modified or the affected attribute excluded from the claims made on the model.

A run will be accepted into the analysis when it was executed within the tolerances of SOP-2003 for the parameters not being varied and when the harvest samples met the acceptance criteria of the analytical methods. A run that fails either condition will be recorded as a deviation, investigated, and either repeated or dispositioned with a documented impact assessment. A study whose design is invalidated by a deviation will be re-executed in full on material that meets the input specifications for the step, and the invalidated dataset will be retained and referenced in PCR-003 without being analysed.

A response surface model will be considered adequate for prediction when the lack of fit test of §5.5 does not reject at $p < 0.05$ against the pure error from the centre points, when the predicted coefficient of determination tracks the adjusted coefficient, and when the residual diagnostics show no structure in the residuals and no departure from normality. A model that fails any of these will not be used to define an operating region or a range. It will be reduced, or the design augmented, and the revised model reported with the reason for the revision. No numeric threshold for the coefficient of determination is set in this plan, because the value that constitutes adequacy depends on the noise of the response, and the lack of fit test measures adequacy against that noise directly.

An effect will be called significant when its term is significant at $p < 0.05$. Significance alone does not make an effect important. An effect will be called practically important when the predicted change in the response across the characterization range is large relative to the width of the acceptance criterion for that attribute in Table 4.1. An effect that is significant but not practically important will be reported as such, with both quantities given.

An operating region will be declared over the well controlled parameters where the response surface models predict that every modelled attribute in Table 4.1 meets its acceptance criterion. The region will be declared only over the parameters that were varied in the response surface design, and only within the characterization ranges of

Table 6.1. The normal operating ranges must lie inside the declared region. If they do not, the operating region will be reported as it is found and the normal operating range narrowed until it fits inside.

A proven acceptable range will be established for each parameter against each attribute whose fitted model contains a significant term in that parameter, by the analyses set out in §8. The plan is met for a parameter when such a range is established and the normal operating range lies inside it for every one of those attributes.

Each parameter will be classified under SOP-4001 from the effects demonstrated in these studies and from the risk that routine operation moves it outside the region in which those effects are acceptable (CMC Biotech Working Group 2009). A parameter whose variability affects a critical quality attribute is a critical process parameter or a well controlled critical process parameter, and the distinction between the two rests on the risk of falling outside the design space rather than on the size of the effect. A parameter that affects process performance without affecting a quality attribute is a key process parameter. A parameter with no demonstrated effect on either is a general process parameter.

Capability will be reported for each attribute as described in §5.5. No numeric capability threshold is set here. The report will state the Cpk for each attribute with its margin to the limit, and will name the attribute whose capability is tightest.

The study as a whole will be considered complete when the scale-down model is qualified, when an adequate model exists for every modelled response or the reason for its absence is reported, when an operating region containing the normal operating ranges is defined, when a proven acceptable range is established for every parameter and every attribute its model shows it changes, and when every parameter of Table 6.1 is classified.

8 Proven acceptable ranges (planned analysis)

A proven acceptable range is the range of a parameter over which the attribute in question is shown by supportive data to remain acceptable (International Council for Harmonisation 2009). This section states how those ranges will be derived, against what criterion, and how they will be reported.

The criterion applied to each response is the criterion that the output of this step must meet. For an attribute that the downstream train neither clears nor modifies, that criterion is the drug substance acceptance criterion in Table 4.1, because the level leaving the bioreactor is the level presented at release. For an attribute that the downstream train does clear, an in-process limit applies at this step, derived from the drug substance criterion and the clearance the downstream steps deliver, with a margin held between the two so that the drug substance does not depend on every downstream step delivering its nominal clearance exactly. The criterion used for each response, and which of the two bases it came from, will be stated in PCR-003 beside the range it produced.

Two analyses will be run for each pairing of an attribute with a parameter. The first holds the other factors at the set-points in Table 6.1 and scans the parameter of interest across the full characterization range, evaluating the fitted response-surface model at 81 grid points. The second propagates the variation of the other factors, drawing 2000 sets of settings from within their normal operating ranges at each of the same grid points and evaluating the fitted model at each draw. The first analysis answers what the parameter does on its own. The second lets the other factors move as they move in routine operation, so the interaction terms of the fitted model contribute to the predicted level at every grid point.

In both analyses the proven acceptable range is the contiguous interval of the parameter, containing the set-point, over which the criterion is met. An interval that is met at the set-point but broken elsewhere in the characterization range will be reported as the contiguous interval, and the break will be described where it occurs. Where the criterion is met across the whole characterization range, the proven acceptable range is that range, and its edges are the edges of the study.

Each proven acceptable range will be reported in a table, per attribute and per parameter, giving the characterization range, the criterion, the range obtained at the set-points and the range obtained under normal operating range propagation. For each modelled attribute, the parameter with the largest main effect on it will also be plotted against it, with the acceptable region shaded green, the set-point marked and the normal operating range marked. The plot shows how far the predicted level sits from the criterion at the set-point and at each edge of the normal operating range.

A proven acceptable range is univariate in the parameter it names. The operating region declared under §7 is the multivariate region over which every modelled attribute is acceptable at once, and it is not the product of the individual proven acceptable ranges. Where the two differ, the difference is the effect of the interactions, and the report will say so.

9 Data management and integrity

All data generated under this plan will be recorded so that they are attributable, legible, contemporaneous, original and accurate, and so that they remain complete, consistent, enduring and available. Run records will be captured in the validated electronic batch record system, and analytical results will be reported from the validated methods of Table 5.1 with their audit trail intact. Raw instrument data will be retained in the form in which the instrument produced them.

The designs will be generated, randomized and analysed under SOP-1002. The design file, the raw response data and the analysis script for each design will be retained together, so that every table and figure in PCR-003 can be traced back to the runs that produced it. Any exclusion of a run from an analysis will be documented with its justification before the analysis is re-run.

Instruments used to set or measure a studied parameter will have their calibration verified before execution and confirmed after it. A calibration failure found after execution will be treated as a deviation, and the impact on the runs performed since the last successful verification will be assessed and reported.

All data and analyses will be subject to second person review before they are used in PCR-003. The reviewer will confirm that the recorded settings match the design, that the reported results match the raw data, and that the analysis follows §5.5 of this plan.

10 Roles and responsibilities

The functions in Table 10.1 carry the responsibilities listed against them. The approval of this plan by those functions is recorded in Table 1.

Table 10.1: Functions and their responsibilities for this study.

Function	Responsibility
Process Development / MSAT	Owns this plan. Executes the studies on the scale-down model, analyses the data and writes PCR-003.
Manufacturing Science	Confirms that the scale-down conditions and the set-points represent commercial operation, and reviews the operating region against the control capability of the commercial equipment.
Analytical Development	Provides the validated methods, releases the analytical data, and assesses method performance where a result falls near an acceptance criterion.
Statistics	Approves the designs before execution, performs or reviews the model fitting and the capability simulation, and confirms model adequacy against §7.
Quality Assurance	Approves this plan and PCR-003, and dispositions any deviation raised during execution.

11 Deliverables and schedule

The deliverable of this plan is PCR-003, the process characterization report for the production bioreactor. It will report the executed designs, the fitted models and their adequacy, the operating region, the proven acceptable ranges, the capability estimate, the classification of every parameter in Table 6.1 and any deviation raised during execution. PCR-003 rolls up into PCMR-001, which consolidates the campaign across the whole drug substance train.

Table 11.1 gives the sequence of activities. The screening design cannot be executed until the scale-down model is qualified, and the parameters of the response surface design are chosen from the screening result, so the phases run in the order shown. No dates are set in this plan. The schedule is held in the campaign plan under PCMP-001.

Table 11.1: Planned phases and their outputs.

Phase	Activity	Output
1	Qualification of the scale-down bioreactor model against at-scale data	Qualification record, referenced in PCR-003
2	Execution of the screening design	Screening dataset and effect estimates
3	Execution of the response surface design	Response surface dataset and fitted models
4	Execution of the univariate assessments	Univariate dataset
5	Analysis, capability simulation and parameter classification	Operating region, proven acceptable ranges, parameter classification
6	Reporting	PCR-003

Confirmation of the ranges at commercial scale is not a deliverable of this plan. It will be performed during the process performance qualification campaign of 5 batches at the receiving site, under the plan referenced in PTP-001 (U.S. Food and Drug Administration 2011).

12 Risks and assumptions

The principal risk to the study is that the scale-down model does not represent the commercial vessel for one of the attributes it is used to claim. The qualification of §5.1 addresses that risk for the attributes it compares, and the criterion in §7 is what the model must meet. The risk that remains after qualification is that the model represents mean performance without representing the variability of the commercial vessel, and it is managed by confirming the ranges at commercial scale during Stage 2.

Material supply is the second risk. The studies require vials of the working cell bank, and lots of basal medium and feed sufficient to execute both designs. A change of medium or feed lot within a design would add a source of variation the design does not model, so lot identity will be recorded with each run and lots will be reserved for the campaign where the quantity allows.

The third risk is analytical. A response measured near the edge of its acceptance criterion carries the precision of its method into every conclusion drawn about it, and an effect smaller than that precision cannot be resolved by any number of runs at this design size. The consequence is stated where it arises in §5.3 and will be stated again in PCR-003 wherever a range is set by a boundary that the method cannot resolve sharply.

The plan rests on three assumptions. It assumes that the platform experience described in §4.1 transfers to A-Mab, which is supported by the shared cell line construction, medium, feed strategy and control scheme, and which the qualification runs test directly. It assumes that the parameter list in Table 6.1 is complete for this step, which RA-001 established and which a univariate result inconsistent with the mechanism would call into question. It assumes that the attributes in Table 4.1 are the attributes the culture conditions change, and that the attributes the purification train removes are characterized at the steps that remove them.

13 Approvals

This plan will be executed only after it has been approved by the functions recorded in Table 1. Any change to the designs, the ranges, the analyses or the acceptance criteria after approval will be documented as an amendment, approved by the same functions, and reported in PCR-003 with the reason for the change.

14 Appendices

14.1 Appendix A. Planned screening design matrix

Table 14.1 gives the coded settings of the 19 planned screening runs, in the letter codes of Table 6.2. The order shown is the design order, and execution follows the randomized order generated under SOP-1002. No response column appears, because no run has been executed.

Table 14.1: Planned screening design matrix, coded settings.

Run	Type	A	B	C	D	E
1	factorial	-1	-1	-1	-1	1
2	factorial	-1	-1	-1	1	-1
3	factorial	-1	-1	1	-1	-1
4	factorial	-1	-1	1	1	1
5	factorial	-1	1	-1	-1	-1
6	factorial	-1	1	-1	1	1
7	factorial	-1	1	1	-1	1
8	factorial	-1	1	1	1	-1
9	factorial	1	-1	-1	-1	-1
10	factorial	1	-1	-1	1	1
11	factorial	1	-1	1	-1	1
12	factorial	1	-1	1	1	-1
13	factorial	1	1	-1	-1	1
14	factorial	1	1	-1	1	-1
15	factorial	1	1	1	-1	-1
16	factorial	1	1	1	1	1
17	center	0	0	0	0	0
18	center	0	0	0	0	0
19	center	0	0	0	0	0

14.2 Appendix B. Planned response-surface design matrix

Table 14.2 gives the coded settings of the 28 planned response surface runs, in the letter codes of Table 6.2. The axial runs are set at -1 and $+1$, which is what makes the design face centred. The order shown is the design order, and execution follows the randomized order generated under SOP-1002.

Table 14.2: Planned response surface design matrix, coded settings.

Run	Type	A	B	E	C
1	factorial	-1	-1	-1	-1
2	factorial	-1	-1	-1	1
3	factorial	-1	-1	1	-1
4	factorial	-1	-1	1	1
5	factorial	-1	1	-1	-1
6	factorial	-1	1	-1	1
7	factorial	-1	1	1	-1
8	factorial	-1	1	1	1
9	factorial	1	-1	-1	-1
10	factorial	1	-1	-1	1
11	factorial	1	-1	1	-1
12	factorial	1	-1	1	1
13	factorial	1	1	-1	-1
14	factorial	1	1	-1	1
15	factorial	1	1	1	-1
16	factorial	1	1	1	1
17	axial	-1	0	0	0
18	axial	1	0	0	0
19	axial	0	-1	0	0
20	axial	0	1	0	0
21	axial	0	0	-1	0
22	axial	0	0	1	0
23	axial	0	0	0	-1
24	axial	0	0	0	1
25	center	0	0	0	0
26	center	0	0	0	0
27	center	0	0	0	0
28	center	0	0	0	0

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